

Original papers

Automated detection and analysis of social behaviors among preweaning piglets using key point-based spatial and temporal features

Haiming Gan^{a,b,d}, Mingqiang Ou^a, Endai Huang^c, Chengguo Xu^a, Shiqing Li^a, Jiping Li^a, Kai Liu^{b,*}, Yueju Xue^{a,d,e,*}

^a College of Electronic Engineering, South China Agricultural University, Guangzhou 510642, Guangdong, China

^b Department of Infectious Diseases and Public Health, Jockey Club College of Veterinary Medicine and Life Sciences, City University of Hong Kong, Hong Kong Special Administrative Region

^c Department of Computer Science, City University of Hong Kong, Hong Kong Special Administrative Region

^d Guangdong Laboratory for Lingnan Modern Agriculture, Guangzhou 510642, Guangdong, China

^e Guangdong Engineering Research Center for Datamation of Modern Pig Production, Guangzhou 510642, China

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ABSTRACT

In the pig industry, the social behaviors of preweaning piglets are critical indicators of their liveability, growth, health, and welfare status. In this study, we developed a novel method based on convolutional neural networks (CNNs) that extracts high-quality spatiotemporal features to detect different types of social behaviors among preweaning piglets: snout-snout and snout-body social nosing, and snout-snout and snout-body aggressive/playing behavior. In our method, piglets were first detected and individually tracked using an integrated CNN-based network. Upon piglet detection, the key points of the piglet body were detected using a key point detector and a 1-second-tracking-unit strategy was adopted to reduce the influence of tracking errors and missing the detection of piglets. To detect suspected social behaviors in a tracking unit, we constructed a self-adaptive spatial affinity kernel function to represent the relationships between piglets, and pairwise piglets with spatial affinity higher than 0.7 were kept for further analysis, while the others were filtered out. Thereafter, high-quality spatiotemporal features consisting of the piglets' spatial affinity and the corresponding linear motion and angular motion intensities were extracted and input into a support vector machine to hierarchically detect and classify piglet social behaviors. In addition, the proportions of the four detected social behaviors associated with the interaction points were analyzed in short video clips and a long video that covered the whole daytime. Our method achieved a favorable detection performance with a recall of 0.9687, a precision of 0.9309, and an F_1 of 0.9494 in the short video clips, and a recall of 0.9576, a precision of 0.9187, and an F_1 of 0.9377 in the 8-h long video episode. This shows the feasibility of the detection of social behaviors for piglets using CNNs.

1. Introduction

The social behaviors of preweaning piglets are critical indicators of their liveability, growth, health, and welfare status, which are major societal and economic concerns of the pig industry (Ison et al., 2017). In general, the social behaviors of preweaning piglets, also known as interactive behaviors, can be classified into three major categories: social playing, social nosing, and aggressive behavior (Van et al., 2014; Verdon et al., 2019). Social playing is a potential indicator of good environmental conditions for piglets (Held et al., 2011; Lawrence, 1987). Social nosing is an affiliative behavior contributing to piglet

growth (Camerlink et al., 2012; 2013). In contrast, aggressive behaviors of pigs usually result in stress, injuries and economic losses (Clark et al., 2013). Therefore, automated detection and analysis of the social behaviors of piglets are beneficial to information management of their health, welfare, and productivity (McFarlane et al., 1995), which can help stockmen perform targeted interventions to enhance treatment effectiveness, reduce adverse impacts on welfare, and promote sustainable piglet production (Holyoake et al., 1996).

Computer vision, a noncontact and low-cost method, has been widely used in pig behavior recognition (Valletta et al., 2017). In general, simple interactive behaviors of animals can usually be recognized from single images according to the spatial relationship among the

* Corresponding authors at: College of Electronic Engineering, South China Agricultural University, Guangzhou 510642, Guangdong, China (Y. Xue).
E-mail address: kailiu@cityu.edu.hk (K. Liu).

Nomenclature			
<i>Symbols</i>		AM	Angular motion
bl	Average body length of all piglets in a pen	CNN	Convolutional neural network
C_t	Central positions of the detected piglets from I_t	DHSVM	Descendant hierarchical support vector machine
F_1	F_1 -measure	FPN	Feature pyramid network
I_t	An image at timestamp t	FP	False positive
In	Sampling interval in the training association network	FN	False negative
$M_{t,t+n}$	Total motion of a piglet between time step t and $t + n$	HRNet	High-resolution network
n	Sampling interval in a tracking unit	IDS	Identity switch
$Precision_k$	Precision for key point prediction	ID	Identity
P_s	Key point of a piglet's snout	IDF ₁	Identity F_1 score
P_h	Key point of a piglet's hip	IoU	Intersection over union
$Precision_s$	Precision for social behavior detection	LM	Linear motion
$Recall_k$	Recall for key point prediction	MI	Motion index that describes the motion of a pair of piglets
$Recall_s$	Recall for social behavior detection	MOT	Multiple object tracking
S_s	Snout-snout social behavior	MOTA	Multiple objects tracking accuracy
S_b	Snout-body social behavior	OPTN	On-line piglet tracking network
t	Time step	RPN	Region proposal network
<i>Vect</i>	Vector	SA	Spatial affinity
θ	Rotation angle of a piglet body	SVM	Support vector machine
<i>Abbreviations</i>		TP	True positive
AP	Average precision	TN	True negative

behavior performers. For example, Tsai et al. (2014) achieved the automated recognition of mating events for cattle based on the length of a circumscribed rectangle of cattle, and Nasirahmadi et al. (2016) detected mounting behaviors between pigs according to the pixel area of an ellipse fitted to the pigs. However, the recognition of interactive behaviors by using only geometrical features (e.g., position, distance, orientation, length, and area) may fail in more complex interactive behaviors, especially when motion features (e.g., acceleration) dominate the behavior expression.

The automated recognition of pig aggressive behavior, an intricate interactive behavior, has been explored by Oczak et al. (2014), Chen et al. (2017), Viazzi et al. (2014), Li et al. (2020), and Lee et al. (2016). In their studies, not only geometrical features (e.g., distance between pigs) but also motion features (e.g., speed, acceleration, and the number of moving pixels) were used to characterize aggressive behavior. The rapid movement characteristic of aggressive behavior demonstrates the indispensability of temporal information in recognition. In addition, the pigs in these studies were large in size and white in color with clean and simple backgrounds, indicating the ease of pig detection using conventional image processing technologies, e.g., Otsu threshold segmentation (Zhang, et al., 2008). In addition, in these studies, the pigs were detected as whole objects without distinguishing different body parts. Thus, interaction points (i.e., body parts with contact) were not analyzed. Pig social playing is highly similar to aggressive behavior (Blackshaw et al., 1997; Camerlink et al., 2016; Held and Špinka, 2011) and has rarely been reported in automated recognition based on computer vision.

Nosing behavior mainly consists of snout-to-snout contact, nosing the body, and nosing the tail. Compared to aggressive or playing behavior, nosing behavior is generally performed at a milder intensity. Indeed, almost all interactions between pigs are either preceded or followed by significant associations with nosing a certain body region (Camerlink and Turner, 2013). In particular, snout-to-snout contact tends to be a positive and affiliative nosing behavior, while nosing the tail is sometimes regarded as a precursor to harmful behavior, e.g., tail biting (Beattie et al., 2005; Breuer et al., 2003; Camerlink and Turner, 2013; Weerd et al., 2005). Thus it is of significant importance to distinguish snout-to-snout and snout-to-tail social behaviors and analyze the interaction points while detecting nosing behavior. Nevertheless, the

automated detection of pig social nosing behavior with high accuracy has rarely been achieved. For instance, in previous studies, a certain body part of a pig was usually searched in accordance with a set of geometrical rules, e.g., a special radian for the hip of a pig (Yang et al., 2018a,b). Because a piglet is not a rigid object and the geometrical features vary from piglet to piglet and from posture to posture, localizing a body part based on special geometrical rules may not be a one-size-fits-all method. The rough approximation of body part detection is difficult even when partial occlusion occurs, e.g., when piglets are partially occluded by the sow.

Convolutional neural networks (CNNs) act directly on raw inputs, automating the process of feature extraction (Jiang et al., 2019; Lorbach et al., 2018), and have been widely applied in animal behavior recognition. The recognition of pig feeding and foraging behaviors (Alameer et al., 2020) has been achieved by using GoogLeNet-like CNNs. Yang et al. (2018a) used a Faster R-CNN architecture to locate and identify individual pigs and then extracted the feeding area occupation rate to recognize feeding behavior. Zhu et al. (2020) and Zheng et al. (2018) both adopted proposal-based CNNs to classify sow postures. In the abovementioned studies, the pig behaviors were simple and humans can distinguish the behavior directly based on a single image. However, the social interactions of piglets before weaning are more complicated and usually involve vast quantities of continuous body contacts. Therefore, frame-level recognition and classification often fail due to a lack of temporal information under this condition (Chen et al., 2020).

To automate the detection of social behaviors among preweaning piglets, a computer vision-based method was proposed to hierarchically detect and classify social nosing and aggressive/playing behaviors. In our study, piglets were first detected and tracked by an integrated CNN. Upon piglet detection, piglet body key points were detected precisely by using a CNN key point detector. Then, both spatial and motion analysis techniques were used to detect and classify the abovementioned behaviors. We conducted experimental tests to validate the effectiveness of the proposed detection method with the ultimate goal of developing a computer vision-based piglet social behavior detection method for the continuous improvement of piglets' growth, health and welfare.

2. Materials

2.1. Animals and experimental installations

This experiment was conducted on a commercial sow farm located in Foshan city, China. A total of 10 Meihua sows, a regional heritage pig breed, individually raised in open floor farrowing pens along with their litters were used in the study. The farrowing pens, with dimensions of approximately 3.8×2.0 m and with a full concrete floor where the sow and piglets were free to roam, contained one sow and 8 ~ 12 piglets whose postpartum day varied from 2 to 21. The sows had an uneven color in appearance, while almost all of the piglets were in black with a few of them mottled (Fig. 10).

An overhead camera (DS-2CD1321D-I, Hikvision®, China) was fixed above each pen in the central position at heights ranging from 2.4 to 2.7 m, permitting a top view of the whole pen. Time-series video data were collected from 7:00 to 18:00 at a frame rate of 10 fps and at a resolution of 1920×1080 pixels. To reduce the computational cost in the subsequent experiments, the frame rate was adjusted to 5 frame/s by removing every other frame, and the images from the videos were resized to 960×540 pixels after recording.

2.2. Definitions of social behaviors and key points

According to the studies conducted by Camerlink and Turner, (2013) and Weller et al., (2019), most of the social behaviors of piglets occur when piglets are not suckling teats or massaging udders. Considering that the interactions between piglets when they gather tightly and engage in suckling the teats or massaging the udders of sows are complicated, even when manually observed, piglet social behaviors during sow nursing were not considered in this study. As such, piglet contacts during sow nursing were eliminated using the sow nursing behavior detection method developed in our previous study (Yang et al., 2018a).

To characterize the motion and the piglet interaction points, the body parts of the snout and hip were selected and denoted P_s and P_h , respectively. As it is rather difficult to separate social playing and aggressive behavior by human observation (Blackshaw et al., 1997; Camerlink et al., 2016; Held and Špinka, 2011), unless otherwise specified, social playing is not distinguished from aggressive behavior in our

paper. The distributions of all key points are depicted in Fig. 1, and the descriptions of all key points and behaviors are given in Table 1. Specifically, a social behavior that is performed by snout-snout interaction is considered snout-snout social behavior, e.g., snout-snout social nosing. Otherwise, it is a snout-body type of interaction (Camerlink et al., 2016).

2.3. Data allocation and labeling procedure

The collected data were divided into two sections: a training dataset and a test dataset. First, the 10 pens were shuffled randomly, and the first seven pens that were drawn were selected as the training dataset to train the detection head, association head and key point detector (see Sect. 3.2). The remaining three pens were used as the test dataset to evaluate the results of piglet detection, key point detection, and piglet tracking. More information on the dataset collection is given in Table 2. Videos from each pen were trimmed to 30 s-footage video clips, and 10 clips were selected from each pen. In addition, an 8-h video episode from one of the three test pens was reserved for further testing. In the long video episode, piglets were performing active social behaviors in the morning, at noon, and in the afternoon, and the sow was nursing occasionally. Finally, there were a total of 100 video clips (70 for training and 30 for testing) and one long video episode. In the selected videos, piglets exhibited nosing or aggressive/playing behavior; chasing, hustling or sleeping; adhesion between piglets; and occlusion from the sow.

The labeling bounding box and the key point for each piglet were processed via Labelme software (Fig. 1). The ID number of each piglet

Table 1
Descriptions of all key points and behaviors.

Key points/ behavior	Description
P_s	Snout of a piglet.
P_h	Hip of a piglet.
Social nosing behavior	Piglet touching or sniffing any part of the head or nose of another piglet. Snout movements may be observed.
Social playing behavior	Nudging or pushing, playing fighting.
Aggressive behavior	Forceful fighting, pushing with the head or biting littermates in a violent manner.



Fig. 1. Ground-truth bounding box and key points in a tracking unit. Only the 1st, 3rd, and 5th frames are displayed for brevity.

Table 2
Dataset collection.

Collection time	Piglet postpartum day	Number of pens for training	Number of pens for test
9:00 ~ 18:00, April 25–29, 2017	2 ~ 30	5	2
8:00 ~ 18:00, September 5–6, 2018	6, 14	2	1

was annotated simultaneously. The ID number in each pen began with “1” and ended up with the total number of piglets in the pen. We displayed the ID number of each piglet and identified the pairwise piglets **in contact**. We first equally segmented the video episodes into independent 1-second (5-frame) video clips starting from the first frame. According to the ethogram in Table 1, we scored the pairwise piglets into noninteraction, social nosing, or aggressive/playing behavior, and we recorded the video names, IDs of pairwise piglets, and timing (video sequence number) of the social behaviors. One pair of piglets performing social behavior lasting for 5 frames was considered one positive sample (social nosing or aggressive/playing behavior). One 5-frame video clip may contain multiple social behavior events. Ground truth piglet social behaviors of video clips were labeled under the guidance of experts in animal behavior.

3. Methods

3.1. Overview of the on-line tracking method

To harness the power of CNNs, we developed a deep learning network system mainly composed of three convolutional neural networks: a piglet detection head, a key point detector, and a piglet association head. The workflow of the proposed method is shown in Fig. 2. Piglets were first detected and tracked with an integrated CNN-based network. Upon piglet detection, piglet body key points (i.e., snout and hip) were then detected precisely by using a CNN-based key point detector. To represent the spatial relationship between piglets, we constructed a self-adaptive spatial affinity kernel function. According to the spatial affinity estimation, suspected social behaviors in a tracking unit were detected. Finally, high-quality spatiotemporal features containing the intensity of the linear motion and angular motion of piglets were extracted and input into a support vector machine to hierarchically detect and classify piglet social behaviors.

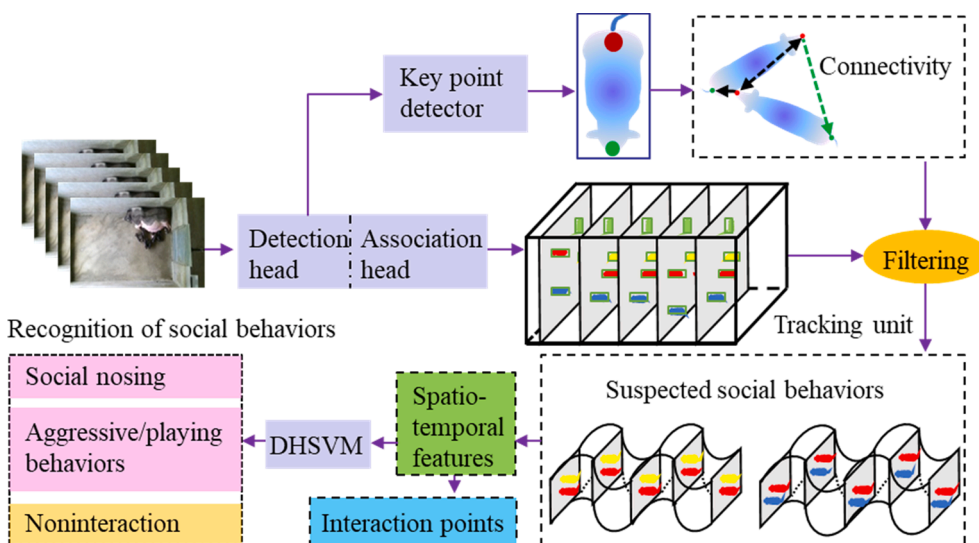


Fig. 2. Workflow of the detection of piglet social behaviors. The key point detector detected piglet snout (green solid point) and hip (red solid point) key points. Low spatial affinity piglet pairs were filtered out to obtain the suspected social behaviors. The spatiotemporal features of piglet pairs in a 5-frame (1 s) tracking unit were extracted and input into a support vector machine to hierarchically detect and classify piglet social behaviors. DHSVM stands for descendant hierarchical support vector machine. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

3.2. Cnns for object detection and tracking and key point detection

Piglet detection and tracking. To extract the spatiotemporal features of the social behaviors of pairwise piglets, following the tracking-by-detection paradigm, we proposed a completely integrated CNN-based online piglet tracking network (OPTN) that performed piglet detection and tracking simultaneously (Gan, et al., 2021). The integration of the piglet detection head and association head is shown in Fig. 3. Given ResNet-101 and the feature pyramid network (FPN) as the base network, OPTN detects piglets with a proposal-based subnetwork and the associated detections of piglets (bounding boxes) across frames mainly by using CNN-based features, which was inspired by the *single shot tracker* (Sun et al., 2019). OPTN is robust against dim lighting conditions and occlusion since we adopted a distance-based strategy to recover identities that were missed or occluded. The distance-based strategy is a piglet tracking method where a missed piglet can be associated with its previous position according to distance.

However, missed detections are always unavoidable due to the complexity of piglets' behavior in the litter and of challenging scenes, which increases the risk of losing the object or introducing tracking errors. Considering that a 'social behavioral event' is classified as such when the piglets engaged in that behavior for at least 1 s, each video episode was segmented into a number of 1-second clips that incorporated 5 video frames to obtain a stable tracking trajectory of pairwise piglets (Liu et al., 2020; Oczak et al., 2014). Each quantity of 1-s clips was termed a *tracking unit* in our study. Consequently, each tracking unit was processed independently, and each trajectory in it was initialized at the first frame of each tracking unit and ended at the last frame of each tracking unit.

Key point detection. Compared to sows, piglets are relatively small in the images and have a purer color appearance, which increases the difficulties in detecting the body key points of the piglets. Inspired by the deep high-resolution network (HRNet) (Ke et al., 2019), we used a multiscale convolutional network to detect key points for piglets. The multiscale convolutional network consists of two stems of strided convolutions ($\times 2$ and $\times 4$) decreasing the resolution and a main stem outputting 5 feature maps with the same resolution ($\times 1$) as its input feature maps. The network structure of the key point detector is shown in Fig. 4. The key point positions are chosen and transformed to the full resolution. Each key point location is predicted by adjusting the highest heat value location with a quarter offset in the direction from the highest response to the second highest response.

Our key point detector is a CNN-based network that 1) connects high-to-low resolution subnetworks in parallel rather than in series, as is the

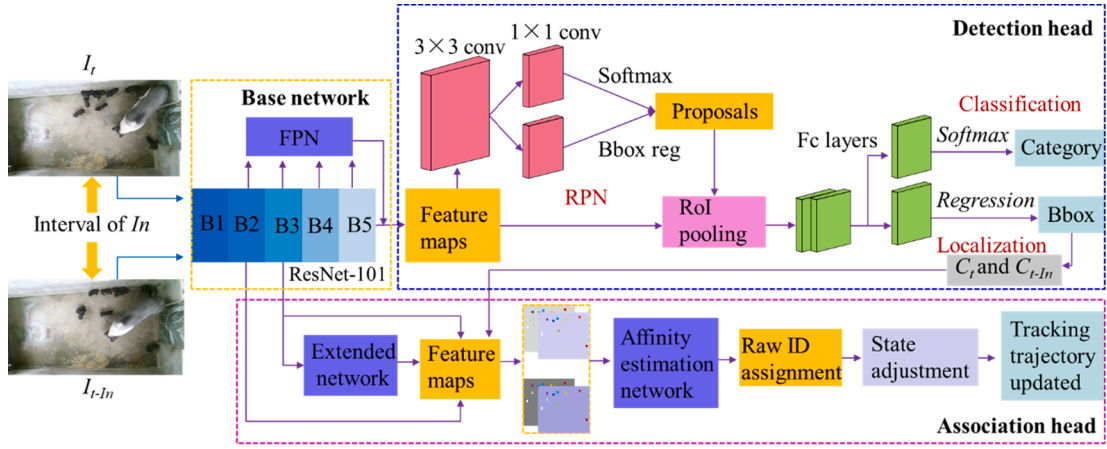


Fig. 3. Overview of the piglet detection and tracking. I_t and I_{t-In} are two images with a sampling interval of In . The base network consists of ResNet-101. The region proposal network (RPN) is in the detection head, and *conv* denotes the convolutional layer, where the numbers before it are the kernel size. C_t and C_{t-In} are the central positions of detected piglets at I_t and I_{t-In} , respectively.

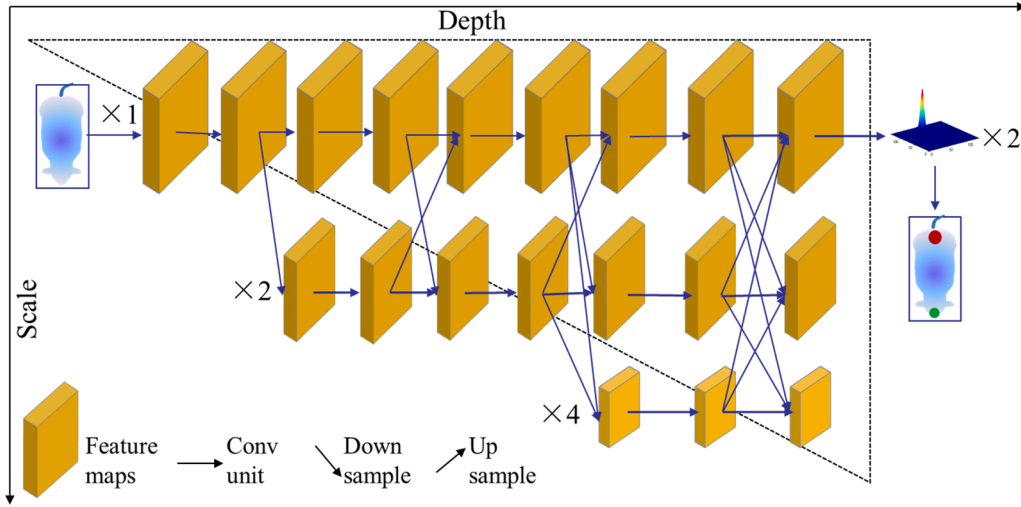


Fig. 4. Network structure of the key point detector. It consists of parallel high-to-low resolution subnetworks with repeated information exchange across multi-resolution subnetworks (multiscale fusion). The horizontal and vertical directions correspond to the depth of the network and the scale of the feature maps, respectively. Two heatmaps are output as the estimation of the positions of two key points.

case in most existing solutions, and 2) performs repeated multiscale fusions to boost the high-resolution representations with the help of the low-resolution representations of the same depth and similar level, and vice versa. The benefit of those network design strategies is that the high resolution is maintained, and accordingly, the predicted heatmap is potentially spatially more precise for small targets (Huang et al., 2020; Ke et al., 2019).

3.3. Detection and classification of suspected social behaviors

Suspected social behaviors detection. An ideal process for detecting social behaviors is to classify the exhaustive permutations of pairwise detections of piglets. However, the computation of exhaustive permutations is time-consuming. Therefore, to reduce the computational cost, only potential social behaviors should be detected within a tracking unit. The snout is the main body part involved in the social behaviors of piglets and is close to the behavior receiver. To manifest the spatial relationship between two piglets, we constructed a self-adaptive kernel function to model the spatial affinity between the two pairwise piglets. We denote the spatial affinity value as SA , which is computed as follows:

$$SA = 3 \times bl / \min \left(\sum_{t=1,3,5} d_{12}^t, \sum_{t=1,3,5} d_{21}^t \right) \quad (1)$$

where d_{12}^t is the sum of Euclidean distances from *piglet*₁ snout to *piglet*₂ snout and hip at time step t , d_{21}^t is the sum of Euclidean distances from *piglet*₂ snout to *piglet*₁ snout and hip at time step t and bl is the average body length of all piglets in a pen. To reduce the influence of missed detections, we selected only three inconsecutive frames (the 1st, 3rd and 5th frames) for SA computation.

Motion description. Nosing behavior can be distinguished from aggressive/playing behavior by using a specified activity index to describe the motion intensity of the behavior. Segmentation-based angular motion, linear motion, and activity index are not robust against object overlapping, occlusion, and incomplete segmentation. Considering the key point prediction's strong robustness against object overlapping, occlusion and variable shooting angles, we proposed a key point-based motion evaluation method for the piglet (Fig. 5). The motion of a piglet is composed of angular motion and linear motion. Since the snout is the most essential body part while considering social behavior, we represent the linear motion of piglets with the linear motion of the snout, instead that of the centroid. Consequently, according

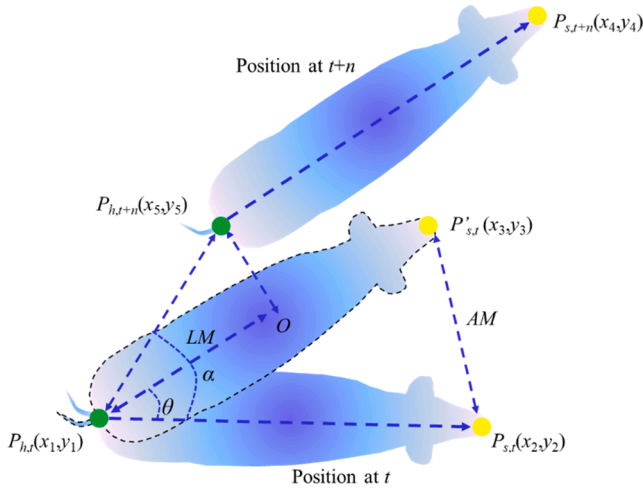


Fig. 5. Key point-based motion description of a piglet. θ denotes the angle of body rotation, α denotes the angle between $P_{h,t}P_{h,t+n}$ and $P_{h,t}P_{s,t}$. AM and LM denote the angular motion of the body and the linear motion of the piglet's snout, respectively. $P_{h,t}(x_1, y_1)$ and $P_{s,t}(x_2, y_2)$ denote two detected key points at time step t and $P_{s,t}(x_3, y_3)$ denotes the corresponding localization along the piglet body orientation at time step $t + n$. $P_{s,t+n}(x_4, y_4)$ denotes the key point of the snout at time step $t + n$. $P_{h,t+n}(x_5, y_5)$ denotes the key point of the hip at time step $t + n$. The piglet outlined by a dotted line is a hypothetical position when only angular motion occurs.

to the notation in Fig. 5, the total motion $M_{t,t+n}$ of a piglet between two video images at time steps t and $t + n$ may be calculated from the following equations:

$$\cos\theta = \frac{(x_2 - x_1)(x_4 - x_5) + (y_2 - y_1)(y_4 - y_5)}{\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \cdot \sqrt{(x_4 - x_5)^2 + (y_4 - y_5)^2}} \quad (2)$$

$$\cos\alpha = \frac{(x_2 - x_1)(x_5 - x_1) + (y_2 - y_1)(y_5 - y_1)}{\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \cdot \sqrt{(x_1 - x_5)^2 + (y_1 - y_5)^2}} \quad (3)$$

$$AM = \sqrt{2((x_2 - x_1)^2 + (y_2 - y_1)^2)(1 - \cos\theta)} / bl \quad (4)$$

$$LM = \sqrt{(x_1 - x_5)^2 + (y_1 - y_5)^2} \times \left(\cos\alpha \cos\theta + \sqrt{(1 - \cos^2\theta)(1 - \cos^2\alpha)} \right) / bl \quad (5)$$

$$M_{t,t+n} = LM + AM \quad (6)$$

where n denotes the sampling interval between two video images for piglet motion evaluation. AM and LM are related to the body size of piglets, and the body size in the video images varies with different video recording heights (from camera to the ground) and the number of postpartum days. To make AM and LM independent of the pig body size, they are divided by body length bl .

Let MI be a motion index that describes the motion of a pair of piglets performing suspected social behavior in a tracking unit:

$$MI = \sum_i (M_{i,t+n}^{(1)} + M_{i,t+n}^{(2)}) \quad (7)$$

where $M_{i,t+n}^{(1)}$ and $M_{i,t+n}^{(2)}$ are the total motion of two pairwise piglets in a tracking unit from time step t to $t + n$, respectively, $t \in \{1, 3, 5\}$ and $n = 2$.

To further describe the motion of interactive behavior, in addition to MI , we also adopted the *average*, *maximum*, *minimum*, and *variance* of the angular and linear intensities to enhance the description of a social behavior pattern (Yang et al., 2018a). To reduce the

influence of missed detections while calculating the motion indexes, the 1st, 3rd and 5th frames ($n = 2$) of a tracking unit are considered for the motion index calculation. This strategy allows a longer temporal interval for the pairwise sampled frames; thus, motion indexes may be more accurate since the proportion of the calculated motion caused by the fluctuation of the key point prediction error is reduced. The detections in the 2nd and 4th frames are used only for better piglet linkage in the tracking phase. The feature extraction from inconsecutive video frames is depicted in Fig. 6. We first calculated the linear motion and angular motion for each of the pairwise piglets individually. Let $LM_{avg}, LM_{max}, LM_{min}$, and LM_{var} be the average, maximum, minimum and variance values, respectively, of the linear motion intensity LM , and $AM_{avg}, AM_{max}, AM_{min}$ and AM_{var} be the average, maximum, minimum and variance values, respectively, of the angular motion intensity AM . Finally, a 10-dimensional vector $Vect [LM_{avg}, LM_{max}, LM_{min}, LM_{var}, AM_{avg}, AM_{max}, AM_{min}, AM_{var}, MI, SA]$ was used as the descriptor for piglet social behavior detection and classification.

Suspected social behavior classification. Upon the detection of suspected social behaviors, a descendant hierarchical support vector machine (DHSVM) hierarchically classified suspected social behaviors (Benabdeslem, 2006). The DHSVM method consists of two major steps: (1) $Vect$ is input into SVM_1 to distinguish interactions from non-interactions, and (2) $Vect$ is input into SVM_2 to classify interactions into aggressive/playing behaviors or social nosing (see Fig. 7).

3.4. Setup

The training and testing of all CNNs were conducted within the Pytorch framework on a single NVIDIA GeForce RTX 2080 TI graphics processor.

ResNet-101 pretrained on the public dataset ImageNet was loaded to initialize the base network for its training. We trained the detection head and association head separately. The total training iteration and initial learning rate for the detection head were $14k$ and 5×10^{-3} and the training iteration and initial learning rate for the association head is 10^5 and 10^{-2} . The learning rate decay factor was 10^{-1} for both trainings. The key point detector was trained with a total of 160 epochs, an initial learning rate of 10^{-3} and a learning rate decay factor of 10^{-1} . In addition, we adopted data augmentation methods (horizontal and vertical flipping, and 180° rotation) to avoid overfitting during the training of the piglet detection head and key point detector.

The multiobject detection evaluation metric AP_{pig} (average precision) was used to evaluate the performance of piglet detection (Liu et al., 2018). AP_{pig} is calculated as follows:

$$Precision_{pig} = \frac{TP_{pig}}{TP_{pig} + FP_{pig}} \quad (8)$$

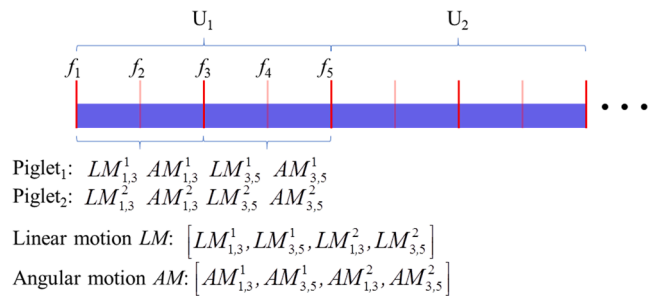


Fig. 6. - Motion feature extraction from a tracking unit. $f_1, f_2, \dots, f_5$ denote 5 video frames in a tracking unit U_1 . LM_{f_1, f_2}^{pi} denotes the linear motion of the pi^{th} piglet between the f_1^{th} and f_2^{th} video frames, while AM_{f_1, f_2}^{pi} denotes the angular motion of pi^{th} piglet between the f_1^{th} and f_2^{th} video frames.

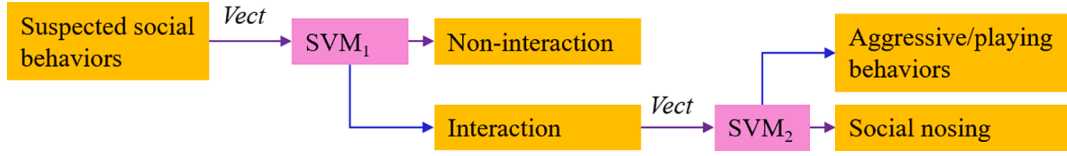


Fig. 7. Hierarchical classification of suspected social behaviors using DHSVM. *Vect* is a vector input into SVM₁ and SVM₂.

$$Recall_{pig} = \frac{TP_{pig}}{TP_{pig} + FN_{pig}} \quad (9)$$

$$AP_{pig,\eta} = \frac{1}{101} \sum_{Recall_{pig}} Precision_{pig}(Recall_{pig}, \eta) \quad (10)$$

where TP_{pig} is the number of detections whose IoUs (intersection over union) over ground truth bounding boxes are greater than or equal to 0.5 while FN_{pig} is the number of missing detections, $Recall_{pig}$ and $Precision_{pig}$ are calculated under an IoU threshold of 0.5, and $Precision_{pig}$ is averaged (AP) by the number of $Recall_{pig} \in [0:0.1:1]$.

Following the *CLEAR MOT* metrics (Bernardin et al., 2008) in the MOT community, we adopted a set of metrics to evaluate the tracking performance of OPTN. In addition to *CLEAR MOT* metrics, the IDF_1 (Ristani et al., 2016) score, was used in our study. The multiple object tracking accuracy (MOTA) and IDF_1 score are the two most important metrics among all the metrics and are calculated as follows:

$$MOTA = 1 - \frac{\sum(FP_{pig} + FN_{pig} + IDS)}{\sum(GT)} \quad (11)$$

$$IDF_1 = \frac{2IDTP}{2IDTP + IDFP + IDFN} \quad (12)$$

where IDS is the number of ID switches between adjacent frames.

For the key point detector, precision $Precision_k$ and recall $Recall_k$ were used to evaluate the key point detection performance (Girdhar et al., 2018; Hosang et al., 2016; Lin et al., 2014; Ronchi et al., 2017). Key point detection is considered correct if the detected key point's Euclidean distance from the corresponding ground truth key point is less than $10\% \times D$, where D is the body length of a piglet from snout to hip (Andriluka et al., 2014; Yang et al., 2013). TP_k , FP_k and FN_k denote the number of key points correctly detected, falsely detected, and missed, respectively.

$$Precision_k = \frac{TP_k}{TP_k + FP_k} \quad (13)$$

$$Recall_k = \frac{TP_k}{TP_k + FN_k} \quad (14)$$

For social behavior detection, $Precision_s$, $Recall_s$ and F_1 (F_1 -measure) (Hripcsak et al., 2005) were adopted to evaluate the performance of the proposed method:

$$Precision_s = \frac{TP_s}{TP_s + FP_s} \quad (15)$$

$$Recall_s = \frac{TP_s}{TP_s + FN_s} \quad (16)$$

$$F_1 = \frac{2Recall_s \cdot Precision_s}{Recall_s + Precision_s} \quad (17)$$

where TP_s , FP_s and FN_s denote the number of social behaviors that were detected correctly, falsely and missed, respectively (Lee et al., 2016). We also used TN_{non} to denote the number of correctly classified noninteractions.

We compared our method with the method developed in previous analogous research in which a CNN was combined with long short-term

memory (LSTM) as a classifier to detect pig tail-biting behavior. Analogous to this method, we applied the pretrained model ResNet-50 (He et al., 2016) as the base network. The total number of training epochs, initial learning rate, dropout rate, and batch size were 50, 10^{-3} (the initial learning rate was searched in the range of $[10^{-2}, 10^{-4}]$), 0.5 (searched in the range of $[0.0, 0.6]$), and 128, respectively, and the solver was Adam. The size of the input image was 224×224 pixels. The CNN combined with an LSTM network was used to classify suspected samples as noninteractions, social nosing, or aggressive/playing behavior.

4. Results

4.1. Performance of piglet detection and tracking and the key point detector

Given using ResNet-101 and an FPN as the base network, piglets were detected with $AP_{pig,0.5}$ and $Recall_{pig}$ of 0.978 and 0.987, respectively. Piglets were tracked with an IDF_1 score of 0.9655 and a MOTA of 0.9704. For the key point detector, the detection results of the two key points are given in Table 3. Piglet key points are detected with a total recall and a total precision of 0.9488 and 0.9710, respectively.

4.2. Performance of the detection of social behavior

The spatial affinities of contact and noncontact pairwise piglets from the training dataset are depicted in Fig. 8. According to the statistical analysis, the spatial affinity threshold between contact and noncontact piglets was approximately 0.8 but was set to 0.7 to incorporate more suspected social behaviors.

Social behavior detection and classification in short video clips

SVM₁ in DHSVM classified suspected social behaviors with a recall of 0.9819, a precision of 0.9656, an F_1 of 0.9737, and a TN_{non} of 867. The final detection and classification results of social behaviors by using DHSVM are given in Table 4. Social nosing was detected with a recall of 0.9877, a precision of 0.9156, and an F_1 of 0.9503. Aggressive/playing behavior was detected with a recall of 0.9320, a precision of 0.9639, and an F_1 of 0.9477. Consequently, piglet social behaviors were detected with a recall of 0.9687, a precision of 0.9309, and an F_1 of 0.9494. The inference speed was 6.35 fps.

In a similar experiment (Liu, et al., 2020), piglet social behaviors were detected with a recall of 0.8949, a precision of 0.8752, and an F_1 of 0.8849. Compared with those obtained by Liu et al. (2020), the recall, precision, F_1 , and inference speed we obtained were 7.38%, 5.57%, 6.45%, and 9.29% higher and the model size was 221 MB smaller in our method.

In the detection results, the proportions of the four different types of social behaviors in short video clips are illustrated in Fig. 9. The proportions of snout-snout and snout-body social nosing (31.73% v.s. 35.94%) and the proportions of snout-snout and snout-body aggressive/

Table 3
Performance of key point detection.

$Recall_k$		$Precision_k$		Total recall	Total precision
P_s	P_h	P_s	P_h		
0.9656	0.9767	0.9433	0.9542	0.9710	0.9488

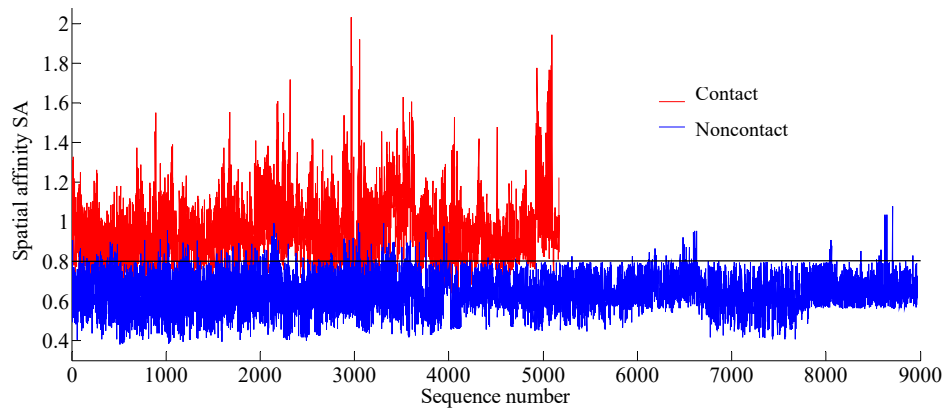


Fig. 8. Spatial affinity between contact piglets (red) and noncontact piglets (blue). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 4

Comparative detection performance in short video clips.

Method	Behaviors	TP_s	FN_s	FP_s	$Recall_s$	$Precision_s$	F_1	TN_{non}	Speed/fps	Model size/MB
Ours	Social nosing	1042	13	96	0.9877	0.9156	0.9503	867	6.35	571
	Aggressive/playing	507	37	19	0.9320	0.9639	0.9477			
	Total	1549	50	115	0.9687	0.9309	0.9494			
Liu et al. (2020)	Social nosing	971	84	150	0.9204	0.8662	0.8925	842	5.81	782
	Aggressive/playing	460	84	54	0.8456	0.8949	0.8700			
	Total	1431	168	204	0.8949	0.8752	0.8849			

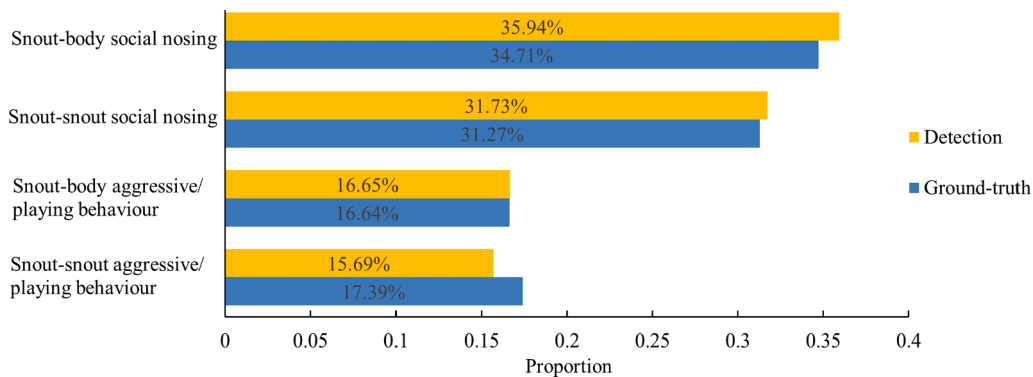


Fig. 9. Time distributions of different types of social behaviors in short video clips. Exemplar cases of social behavior detection under various conditions are given in Fig. 10. Only piglets performing social behaviors have been highlighted with segments.

playing behavior (15.69% v.s. 16.65%) are comparable. In addition, social nosing accounts for 67.67% of social behaviors while aggressive/playing behavior accounts for 32.33%.

Social behavior detection and classification in a long video episode

We tested our method in a long video episode recorded during the daytime. The number of tracking units of social nosing or aggressive/playing behavior was summed per 10-min session (Ott et al., 2014). The detection results and ground-truth data are compared in Fig. 11.

The proportions of different types of social behaviors in a long video episode are given in Fig. 12. The detection proportions of snout-snout and snout-body social nosing are 34.44% and 38.64%, respectively, and the corresponding proportions of snout-snout and snout-body aggressive/playing behavior are 13.29% and 13.64%, respectively. In addition, social nosing accounts for 73.08% of social behaviors, while aggressive/playing behavior accounts for 26.92%.

For the 8-h long video episode, we summarize the social behavior detection performance of our method in Table 5. Social nosing was detected with a recall of 0.9576, a precision of 0.9187, and an F_1 of 0.9377. Aggressive/playing behavior was detected with a recall of

0.9085, a precision of 0.9675, and an F_1 of 0.9371. Consequently, piglet social behaviors were detected with a recall of 0.9434, a precision of 0.9318, and an F_1 of 0.9376. The total TN_{non} was 236.

5. Discussion

In our study, CNNs were used to achieve high-accuracy detection, key point detection and tracking. To the best of our knowledge, this study is the first to report the automated detection of piglets' social behaviors based on a computer vision-based method.

5.1. Main features of the proposed method

Key point-based high-quality spatiotemporal features. The conventional segmentation-based activity indexes developed in Kashiha et al., (2014) and Nasirahmadi et al., (2015) were based on a fitting ellipse that required targets to be precisely segmented without occlusion. These activity indexes have failed to represent animal motion when the body rotation was larger than 90° , as the head and hip were

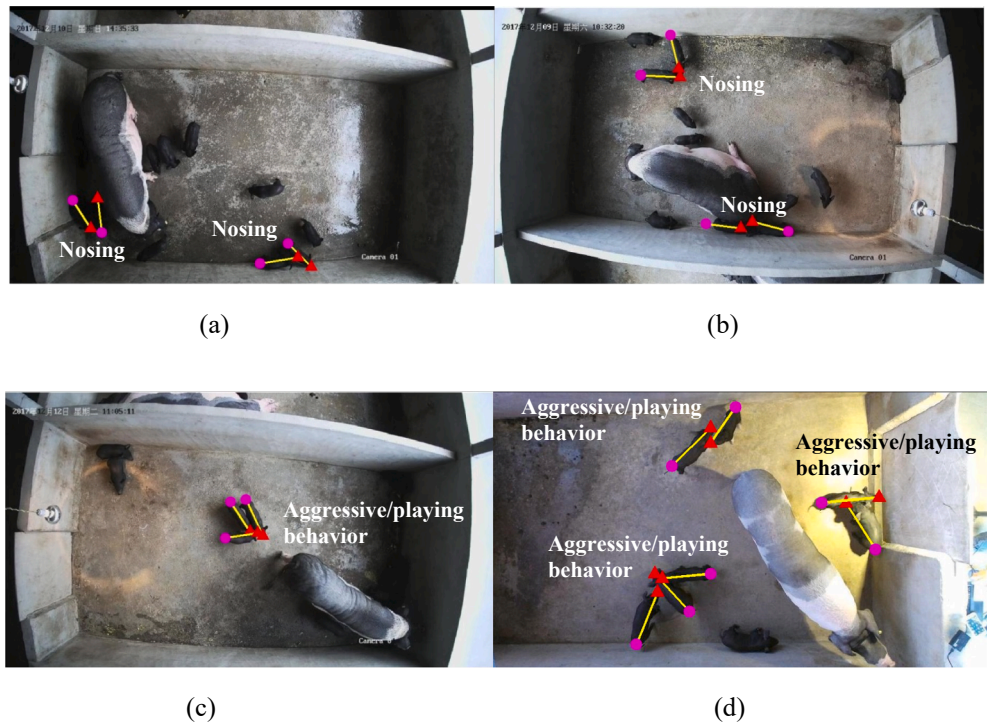


Fig. 10. Social behavior detection results under difficult conditions. The piglets performing social behaviors are highlighted with segments defined by $P_h()$ and $P_s()$. (a) and (b) Detection performance under difficult lighting conditions. (c) An aggressive/playing case that involves three rapidly running piglets. (d) Detection performance under a heat lamp.

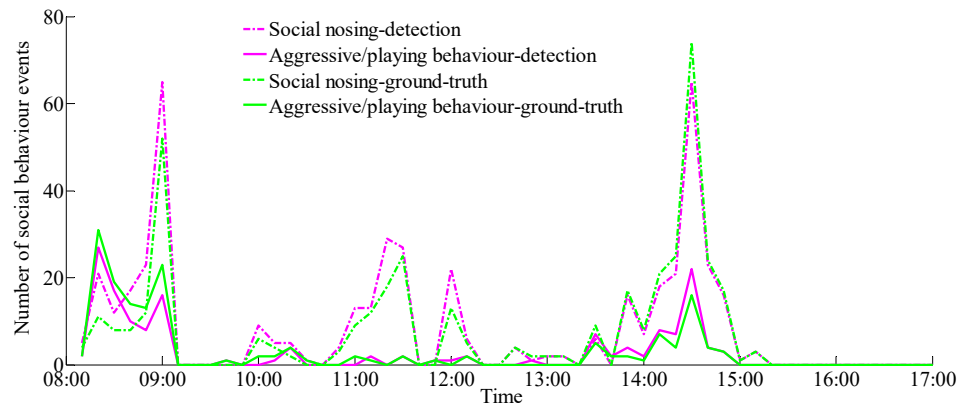


Fig. 11. Detection results and ground-truth data. The number of tracking units of social nosing or aggressive/playing behavior was summed per 10-min session.

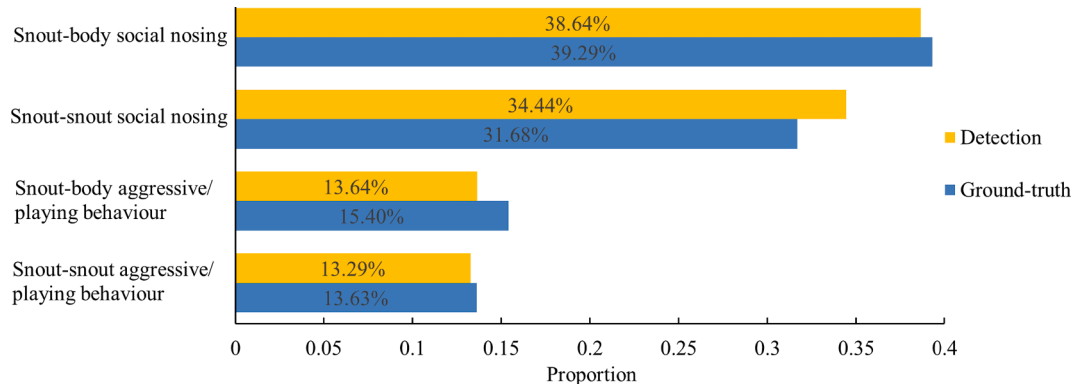


Fig. 12. Time distribution of different types of social behaviors in a long video episode.

Table 5

Detection and classification results of social behaviors in a long video episode.

Social behaviors	TP_s	FN_s	FP_s	$Recall_s$	$Precision_s$	F_1	TN_{non}
Social nosing	384	17	34	0.9576	0.9187	0.9377	236
Aggressive/playing behavior	149	15	5	0.9085	0.9675	0.9371	
Total	533	32	39	0.9434	0.9318	0.9376	

indistinguishable. This rough approximation of total motion is insensitive to minute-scale body movements and thus is not applicable to the detection of social nosing behavior and the distinction between social nosing and aggressive/playing behavior. In addition, the fit of an ellipse is difficult to determine due to the adhesion between piglets since this approach relies on complete piglet body segmentation. To tackle this problem, we proposed a highly precise key point-based method to better characterize the decomposition of piglet body motion.

Our method distinguishes the piglet hip from its snout. Since our key point-based motion index relies on distinguishable piglet body part localization, our method can be applicable for wider angle body rotation. The CNN may tactfully predict occluded key points by inference in our study and thus tackle the occlusion problem, which is unsolved in segmentation-based methods.

In terms of spatial features, we constructed a self-adaptive spatial affinity kernel function to represent the spatial relationship between pairwise piglets. On this basis, we can 1) reduce the computation time to detect suspected social behaviors and 2) use the spatial affinity index, a high-quality spatial feature, as a component of the input vector in the subsequent prediction by using DHSVM. Unlike the temporal features developed by Yang et al. (2018a), which uses optical flow, a pixelwise process that costs vast offline computational resources, our key point-based features may leverage the fast and online parallel computation of a graphics processing unit to improve the inference speed.

Sophisticated social behavior classification. Our method detects not only aggressive behavior but also nosing and playing behavior in a computer vision-based framework. Different types of social behaviors (snout-snout and snout-body social nosing, snout-snout and snout-body aggressive/playing behavior) associated with different body parts were distinguished, which is meaningful in the analysis of piglet behavior. According to the ID relationship between pairwise piglets, it is easy to determine the social behavior that involves three or more performers (Fig. 10 c and d).

Robustness against piglet body deformation and challenging environments. Data-driven and straightforward key point prediction by using a CNN allows our method to be robust against piglet body deformation and partial occlusion, avoiding handcrafted rules for pig body part localization (Yang et al., 2018a). Unlike previous studies on the automated recognition of aggressive behavior (Chen et al., 2017; Liu et al., 2020; Oczak et al., 2014; Viazzi et al., 2014), our study was carried out on preweaning piglets that were small and black with dirty and dim backgrounds and may be occluded by sows, which make object detection difficult. To tackle these problems, we harnessed a CNN to achieve high-accuracy piglet detection and key point detection in challenging environments. To reduce the influence of missed detection and frequent occlusion, we tracked the piglets with a deep learning-based tracking method (that we developed by ourselves) that can adjust false ID initialization, and the 1 s tracking-unit strategy allows our method to be robust against ID errors.

5.2. Detection result analysis

Upon estimating the spatial affinity, the suspected social behaviors mainly consist of cases in which piglets are performing social behaviors or have no interaction (e.g., sleeping). Undoubtedly, sleeping behavior is easy to filter out by using motion features. A piglet passing by another piglet with its snout above or close to the body of the piglet may result in a falsely detected social behavior. From Table 4 and Table 5, we find that

the recall of aggressive/playing behaviors is remarkably lower than that of social nosing. We conjecture that the recall constraint is due to the different levels of difficulties in piglet detection and tracking. Piglets performing aggressive/playing behaviors usually move their bodies more rapidly and frequently, which may cause a range of image problems, e.g., motion blur and severe occlusion. These problems can downgrade the object detection and tracking performance. However, piglets performing social nosing display a milder intensity of motion, which indicates fewer difficulties in object detection and tracking. Our method outperformed that of Liu et al. (2020) by a large margin, as shown in Table 4. We conjecture that the CNN-based features may not be distinguishable enough to separate social nosing and aggressive/playing behavior, while our key point-based method could capture subtle motion differences. The inference speed in our approach is higher for smaller model sizes.

For the detection results of the long video episode as shown in Fig. 11, we find that social behaviors were mainly performed in three time slots, i.e., 8:00 ~ 9:00 in the morning, 11:00 ~ 12:30 (around noon), and 14:00 ~ 15:00 in the afternoon. We went back through the video and found that piglets dispersed when the sow changed posture from lying to standing, after which they actively ran around and interacted more with their littermates (Van et al., 2014; Zheng et al., 2020). As shown in Fig. 11, the curves of the detection data and ground-truth data are close, for both social nosing and aggressive/playing behavior, which also demonstrates a superior detection performance. The proportions in Fig. 12 indicate that social nosing accounts for 73.08% of social behaviors, which is lower than that in study by Van et al. (2014) (89.92%). The nosing behavior in (Van et al., 2014) incorporates the interaction between sows and piglets. Analogous to the detection results in short video clips, the proportions of snout-snout and snout-body social nosing (34.44% vs. 38.64%) and the proportions of snout-snout and snout-body aggressive/playing behavior (13.29% vs. 13.64%) are close. As shown in Fig. 12 and Fig. 9, the proportions of detection data and ground-truth, for both social nosing and aggressive/playing behavior data are also close.

5.3. Limitations

There are three main limitations in our study: 1) Unsustainable tracking causes our method to fail in individual piglet tracking and then long-term individual behavior tracking for piglets. To date, long-term (e.g., 0.5 h or longer) multiple object tracking using computer vision-based methods is still difficult due to the accumulation of false IDs (Li et al., 2015). It is hardly possible to adjust a false ID without human intervention, which is a problem associated with all computer vision-based MOT methods. However, our study focuses on the pen-level frequency of social behavior in a piglet litter. Therefore, the accumulation of false IDs has a marginal effect on the detection performance since we adopted a 1-second-tracking-unit strategy that does not require the IDs of piglets to be identical to those in the initial frame of a video. 2) Our method cannot distinguish aggressive behavior from playing behavior. As mentioned above, it has been difficult to separate social playing and aggressive/fighting behavior by human observation since it takes minutes or longer to observe the reaction of the receiver (Blackshaw et al., 1997; Camerlink et al., 2016; Held and Špinka, 2011). This is a complicated and interesting ethology issue that we will discuss in our future work. 3) Due to space constraints, our method neglects the interactions between piglets and the sow. The interactions between piglets

and sows could be more complicated since it includes more types of interactions, e.g., nursing behavior, and the body of the sow is too large to determine the spatial relationship with piglets. Nonetheless, this can be foreseeably achieved by sow segmentation or other computer vision techniques.

5.4. Implications

Although our study investigated social behaviors in litters of piglets, the interactions between sows and piglets and between piglets and different pen objects (e.g., the nipple drinker and feed trough) remain to be explored by a computer vision-based method. The precise and distinguishable body part localization allows our method to be easily extended to the detection of other piglet behaviors, e.g., social nosing between sows and piglets. Distinguishable body part localization may also contribute to group behavior pattern analysis, e.g., group posture pattern analysis (Nasirahmadi et al., 2015). Interactions without body contact are also important in behavioral studies of pigs, e.g., chasing behavior and vocal interactions among pigs. Piglet movement trajectory features may be extracted by using our tracking method and key point-based trajectory description to detect chasing behavior. As we used a CNN to achieve piglet detection, key point detection and tracking, an integrated network is expected to be designed to make the detection more applicable.

In future research on animal behavior detection applications, long-term statistical analysis based on the social behavior detection results of piglets can be used to analyze the relationship between health status and the social behaviors between piglets. For example, the increasing proportion of snout-snout social nosing is mainly involved in social processes that are likely to be beneficial to the individual or to group cohesion and growth rate (Camerlink and Turner, 2013).

Our dataset was collected from different pens that housed piglets with a varying number of postpartum days under different lighting conditions and during the whole daytime period. For realistic pig production applications, in our future research, we will augment the dataset with a range of breeds of pigs at different growth periods to verify our method and better estimate the transferability of our method to different breeds and environments.

To improve the tracking efficiency, a one-shot tracker (Zhang et al., 2020) which detects and associates pigs simultaneously may be used. A more simplified and integrated framework is also potential in the improvement of the behavior detection efficiency and the reduction of network complexity.

6. Conclusions

In this study, we successfully developed an automated detection method for characterizing the social behaviors of preweaning piglets and our method achieved a favorable detection performance with a recall of 0.9687, a precision of 0.9309 and an F_1 of 0.9494 in short video clips and a recall of 0.9576, a precision of 0.9187, and an F_1 of 0.9377 in the 8-h long video episode. This favorable detection performance demonstrates the feasibility of social behavior detection for piglets using CNNs.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.compag.2021.106357>.

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